AI AND ML FOR MECHANICAL ENGINEERS

Course Code	*<tbd>*		**Examination Scheme**			
Teaching Scheme	3-0-0-0		**Theory**	TA: 20	MSE: 30	ESE: 50
Credits	3		**Laboratory**	ISE: --	ESE: --	--

Course Outcomes: Students will be able to:

1. Demonstrate fundamentals of AI&ML, apply feature extraction and selection techniques
2. Apply machine learning algorithms for classification and regression problems
3. Devise and develop a machine learning model using various steps
4. Explain concepts of reinforced and deep learning
5. Simulate machine learning model in mechanical engineering problems

Introduction: **[7 Hrs]**

History of AI, Comparison of AI with Data Science, Need of AI in Mechanical Engineering, Machine Learning Basics: Reasoning, problem solving, Knowledge representation, Planning, Learning, Perception, Motion and manipulation. Approaches to AI: Cybernetics and brain simulation, Symbolic, Sub-symbolic, Statistical. Approaches to ML: Supervised learning, Unsupervised learning, Reinforcement learning

Development of ML Model: **[8 Hrs]**

Problem identification: Classification, Clustering, Regression, Ranking. Steps in ML modelling: Data Collection, Data pre-processing, Model Selection, Feature Extraction, Feature Selection, Model training (Training, Testing, K-fold Cross Validation), Model evaluation (understanding and interpretation of confusion matrix, Accuracy, Precision, Recall, True positive, false positive etc.), Hyper parameter Tuning, Predictions

Feature Extraction and Selection: **[7 Hrs]**

Feature extraction: Statistical features, Principal Component Analysis. Feature selection: Ranking, Decision tree - Entropy reduction & information gain, Exhaustive, best first, Greedy forward & backward

Classification & Regression Classification: **[8 Hrs]**

Decision tree, Random forest, Naive Bayes, Support vector machine. Regression: Logistic Regression, Support Vector Regression. Regression trees: Decision tree, random forest, K-Means, K-Nearest Neighbour (KNN)

Deep & Reinforced Learning: **[7 Hrs]**

Introduction to Deep learning: Artificial neural networks, Deep neural networks, Convolutional neural network, Recurrent neural network, Long short term memory LSTM, Auto encoder (AE), Dip belief network, Generative adversarial network, Extreme learning machine, Deep residual network. Introduction to reinforced learning: Motivation, background, characteristics, Algorithms for control learning: Criterion of optimality, Brute force, Value function, direct policy search, Model-based algorithms, Associative reinforcement learning, Deep reinforcement learning, Inverse reinforcement learning, Safe Reinforcement Learning, Partially Supervised Reinforcement Learning (PSRL)

Advanced & Recent Techniques: **[8 Hrs]**

Pitfalls of traditional ML models, Data expansion and augmentation, Adversarial Machine learning (AML), Domain generalization and adaptation, Transfer Learning (TL), Explainable AI (XAI), Digital Twin (DT), Multi-models, Multi-agent adaptation, Self-paced learning

Textbooks:

[1] Tom M. Mitchell, “Machine Learning”, McGraw-Hill, 1997

[2] ShaiShalev-Shwartz and Shai Ben-David, “Understanding Machine Learning”, Cambridge University Press, 2017.

[3] Mehryar Mohri, Afshin Rostamizadeh and Ameet Talwalkar, “Foundations of Machine Learning”, MIT Press, 2012

[4] P. Flach, “Machine Learning: The Art and Science of Algorithms that Make Sense of Data. First Edition”, Cambridge University Press, 2012.